Balancing Generative AI Use in Creative Processes: How Teachers and Students Adopt Emerging Technology in an Educational Context

Tore Andre Ringvold, Ingri Strand, Peter Andre Haakonsen and Stephanie Hoebeke
Oslo Metropolitan University
torand@oslomet.no

Abstract

Artificial intelligence (AI) text-to-image generators have quickly become influential in art, design, and media education. While some view it as a fruitful supplement to their creative processes, as it can stimulate idea generation and make image creation quicker and more accessible, others fear that the generators will replace the human creator and eradicate creativity. Furthermore, AI generators are known to reproduce stereotypes, and it can be challenging to control their output. Due to the emergent nature of the technology, there are many studies where the authors themselves or participants have explored using generative technology in specific tasks given by the researchers or in researcher-led educational interventions. Teachers, designers, and artists' perceptions of text-to-image generators are also well studied, but there is less research on how they actually use generative AI in their everyday professional lives. As the technology now has been in common use for a few years, it is time to explore how early adopters have integrated text-to-image generators in their professional practice. We have conducted semi-structured interviews with teachers in Norwegian upper secondary art, design, and media education, strategically chosen due to their advanced use of generative AI. In the interviews, we have asked how they use the technology in their work, both in educational projects with students and in their professional development. We have also asked them to reflect on how they understand and seek to develop their students' AI literacy, as well as the technology's limitations, practical and ethical challenges of educational use of AI. This paper provides insights into how generative AI is used in art, design, and media education, including innovative practices and how the teachers seek to balance its positive and negative aspects in an educational context.

Key Words: Artificial intelligence, Text-to-image generation, Art and design education, AI literacy, Emerging technology

1 INTRODUCTION

As society and technology evolve, education adapts and develops in response. For art, design, and media education to remain relevant, these subjects must equip students with the skills necessary to navigate a changing world. Artificial intelligence is already reshaping society, and educators must engage with its implications, as it stands to redefine how we create, work, and teach. Text-to-image generative technologies have evolved steadily since 2014, gaining widespread attention after the introduction of DALL-E in 2021 (Cetinic & She, 2022). Since then, numerous providers have emerged, supplying an array of standalone services—whether subscription-based or funded through advertising—alongside enhanced functionalities integrated into established software platforms. Such rapid technological progression demands a critical examination of its capabilities and the ways it should be employed in practice.

1.1 Artificial intelligence literacy in education

As AI technologies become increasingly integrated into schools, both teachers and students need to develop a critical awareness of when and how these tools should—and should not—be used. In Norway, county authorities differ in their policies regarding which AI tools are permitted. Approved tools are usually modified chatbot models with strengthened privacy controls. Text-to-image models are not addressed in existing guidelines. Bringing AI into education raises important questions about how teachers make decisions and maintain control over its use. With AI technology advancing rapidly, educators must be prepared to navigate its complexities.

AI literacy (Kong et al., 2024; Chiu, 2025; OECD, 2025) prepares students to critically understand when and how to use AI, knowing its advantages and limitations. It encompasses teaching students to critically engage with AI, including its foundational principles and its practical applications in both educational settings and daily life for problem-solving (Kong et al., 2024). Chiu (2025) argues that AI literacy is interdisciplinary, tightly woven into ten other literacies. In this paper, visual literacy and design literacy are particularly relevant. Students in art, design, and media education learn to interpret and communicate through visual language. The ability to critically assess visual outcomes from AI generators is therefore essential. The connection to design literacy concerns its iterative and human-centred approach as well as a focus on critical thinking and problem solving. In the Norwegian curricula, digital skills are defined as a basic skill. Most subject specific curricula have at least one competency aim including the use of digital technology.

The study has been guided by the overarching research question: *How do teachers balance the challenges and opportunities of using AI text-to-image generators in upper secondary art, design, and media education?*

This has been divided into four sub-questions, where (i-iii) focuses on opportunities and (iv) on challenges:

(i) How do the teachers use generative AI in their teaching?
(ii) How do the teachers seek to develop their students' AI literacy?

(iii) What value do the teachers see in the use of AI text-to-image generators?
(iv) Which challenges and hindrances exist to the use of generative AI?

2 LITERATURE REVIEW

The research on AI text-to-image generators in art, design, and media is characterised by the technology's emergent nature. Several studies draw on researchers' own experimentation (Abdallah & Estévez, 2023; Mikkonen, 2023; Ringvold et al., 2023, 2024), while others involve participants using such tools in various design and artistic processes, including product design (Chen et al., 2019; Liu et al., 2023), furniture design (Chandrasekera et al., 2024), and visual arts (Ko et al., 2023; Lyu et al., 2022). Oppenlaender, Linder, and Silvennoinen (2025) examined participants' ability to assess and develop quality prompts, arguing that prompt engineering constitutes a creative skill.

Within educational research, Vartiainen and Tedre (2023) explored pre-service craft teachers' and teacher educators' understandings and attitudes towards AI text-to-image generators. Participants acknowledged the new opportunities enabled by AI but raised concerns about bias, copyright violations, and lack of transparency. Vartiainen, Tedre, and Jormanainen (2023) arranged hands-on workshops with youth aged 14-16, including exploring and experimenting with the text-to-image generator Midjourney. The students engaged in a free-flowing ideation process but expressed divergent views regarding image ownership. Hutson and Lang (2023) used text-to-image generators into their own digital media course. The students stated that it helped them to structure and visualise ideas but felt that their images shared the same postcard-like expression. Heaton, Low, and Chen (2024) included text-to-image generators in the ideation phase of a painting project with students aged 14–15. Although the students initially expressed concerns related to artistic autonomy and the creative process, they later found that their processes were enriched and that visual exploration with generated images allowed them to engage more deeply with the topic.

Overall, few studies examine how text-to-image generators are used in educational contexts, indicating a research gap. Oppenlaender (2022) conducted an online ethnography focused on prompt-writing. Kalving et al. (2024) investigated how designers use and perceive generative AI, where several participants described using text-to-image generators for ideation, yet also viewed AI use as laziness or as a potential threat to their jobs. In a study on Design and Technology-teachers' adoption of generative AI, McLain, Barrow, and Ndiaye (2024) found that although teachers generally held positive views of the technology, few employed it in innovative pedagogical ways.

3 METHODOLOGY

In this section, we present the methods used for collecting and analysing data, as well as the ethical considerations we have made in planning, conducting, and disseminating the study.

3.1 Semi-structured interviews with teachers

Four informants were purposively sampled (Bryman, 2016). They were identified through inquiries within our professional network, which sought teachers in upper secondary art, design, or media education who could be considered early adopters of AI text-to-image technology. *Early adopters* are defined as those who are among the first to use a new technology (Rogers, 2003). We contacted four teachers meeting these criteria, and all agreed to participate. The informants have diverse backgrounds, with most having professional experience in art, design, or media. All hold pedagogical qualification or teacher education and are currently teaching in upper secondary schools.

Semi-structured interviews were chosen to ensure coverage of predetermined themes while also allowing for follow-up questions (Brinkmann & Kvale, 2015; Bryman, 2016). The interviews were conducted individually and lasted 45-60 minutes. Three to four researchers participated in each, with two researchers present at all sessions. Three interviews were held face-to-face and one via video conference between December 2025 and January 2026.

The teachers were asked about how they use the technology in their professional practice. This included how they aim to develop their students' AI literacy and how they integrate generative AI into creative processes, as well as the technology's limitations, practical and ethical challenges. Finally, they were invited to reflect on text-to-image generators' contribution to educational projects and imagine how the technology could be used in new ways.

3.2 Thematic analysis

The empirical material was analysed through a thematic analysis (Braun & Clarke, 2022). Each author first read the interview transcripts and independently made In Vivo codes (Saldaña, 2016) separately. The subsequent stages of the analysis were conducted collaboratively. Using Microsoft Whiteboard—a digital workspace enabling real-time collaboration—we placed each code on a digital sticky note. In the next step, we grouped similar codes together. Through a process of defining, renaming, and merging (Braun & Clarke, 2022), the codes were developed into themes that characterised how the teachers used AI text-to-image generators, the challenges and hindrances they encounter, how they sought to develop their students' AI literacy, and their understanding of the value of using AI in education.

3.3 Ethical considerations

Before contacting the informants, our plan for handling personal data was approved by Sikt – Norwegian Agency for Shared Services in Education and Research. This approval required us to obtain informed consent from all informants, minimise privacy risks by collecting only necessary data, secure identifiable information through two-factor authentication, and describe informants in a manner that reduces the likelihood of identification (Sikt, undated).

3.4 Disclosure on the use of Generative AI

The authors have conducted the analysis and text production for this paper without the use of Generative AI. The GenAI-tool Microsoft 365 Copilot has been used in copy editing the final version of the text.

4 RESULTS

The structure of this section is guided by the four sub-questions presented in the introduction, where we present a total of 21 themes identified through the analysis.

4.1 The teacher's use of generative AI in their teaching

4.1.1 Getting to know the technology

The teachers wished to introduce students to AI text-to-image generation, aiming to develop an understanding that the technology exists and how it works. In one exercise, students were asked to formulate a prompt describing the teacher. This was followed by a discussion of the generated images and how closely they resembled the teacher.

4.1.2 Inspiration and idea generation

AI text-to-image generators were used to support students' creative processes. Students used AI to generate inspiration images for their projects, rather than searching online, and to explore and refine ideas by experimenting with AI generators.

4.1.3 Producing raw material used as elements in final products

AI-images was not wanted as finished outcomes. Instead, students used AI-generated elements as components within larger compositions, for example by inserting AI-created elements into photographs, modifying specific parts of an image, or assembling multiple AI-generated image or video components into a cohesive whole.

4.1.4 Creating complex or impossible visualisations

The teachers described how students used generative AI to create images that would otherwise be inaccessible or impossible to make—either because existing image databases lacked the necessary material, or because students did not have the equipment, time, or opportunity to access specific locations. One example was a project titled *Under the Surface,* where students created images depicting underwater or underground environments.

4.1.5 Visualising ideas

AI generators were also used to visualise the students' ideas for communication. This included generating visual material to articulate prefigurative thoughts, as well as producing moodboards or storyboards.

4.1.6 Develop critical understanding of images

The teachers also addressed that visual material encountered daily can be misleading and AI-generated, and they aimed to help students develop critical awareness of this. One teacher, for instance, asked students to use AI to fabricate news articles and then challenged them to distinguish them from authentic news articles.

4.2 How the teachers seek to develop their students' AI literacy

4.2.1 Developing a critical understanding of visual outputs

The teachers were concerned with helping students develop a critical understanding of visual outputs, and to distinguish between what is authentic or not, as described above. Informant 1 described spending considerable time with the students reflecting on 'how they are getting tricked'.

4.2.2 Authorships and transparent use of AI

Generative AI actualises a discussion on authorship. Our informants required a transparent use of AI, asking that the students could describe how AI has been used in the process and which elements are generated and which is their own work.

4.2.3 Mastering generative AI and developing skills in prompt engineering

Some of the teachers described teaching students how to master the AI technology, emphasising how the quality of the output is dependent on a clear vision and description, as well as the ability to generate multiple images and evaluate which ones to develop further. Some teachers guided students in using subject-specific terminology and collaboratively developing and assessing their prompts. Taking a different approach, one teacher used the AI-generator Leonardo to upload images and see which prompt could be used to produce a similar result.

4.2.4 Devaluing AI literacy, emphasising visual literacy

Other informants placed less emphasis on AI literacy, with Informant 3 asking, 'is AI literacy really necessary?'. Rather than teaching prompt-engineering, one teacher instructed the students to use Chat GPT to generate prompts. These teachers instead prioritised the development of visual literacy.

4.3 How the teachers value the use of generative AI in teaching

4.3.1 Preparing the students for a future with AI

The teachers regarded the use of generative AI in education as essential for preparing students for the future and preventing them from lagging behind. Informant 3 described his first understanding of text-to-image technology as 'revolutionary in the field'. The media teachers stated that they expect generative AI to be used increasingly in the industry, which requires them to use the technology in education.

4.3.2 Improving the creative process and outcome

Several informants described how text-to-image technology enhanced students' creative outcomes. Generating their own images for ideation or moodboards can increase sense of ownership and improve the communication of design ideas. The teachers also stated that generative AI can enable new artistic expressions and creating "impossible" images.

4.3.3 Making the creative process more efficient

Some teachers also emphasised how AI technology is far more efficient than traditional techniques, allowing for 'a whole sketching process in a few minutes', as Informant 3 stated.

4.3.4 Having fun experiencing the technology

One informant also mentioned that the students enjoy using text-to-image technology and have fun with the experience, as an important value of using the technology.

4.4 Challenges and hindrances to the use of generative AI

4.4.1 Limitations in available tools

The text-to-image generators available to the teachers had limited functionality and poor quality, limiting the students' actions and results. Their choice of AI-generators was restricted by economy and privacy regulations. All the teachers stated that without these limitations, they would use more advanced and more costly alternatives.

4.4.2 Students' capabilities in using AI

The informants described limitations in students' prompting abilities and in their persistence when working toward a desired outcome. One teacher emphasised the importance of encouraging AI use primarily in the second or third year, expressing concern that first-year students should first develop foundational skills in image making and design before incorporating text-to-image generators into their creative processes

4.4.3 Resistance from the students

Among the informants, clear differences emerged between the educational programmes in which the students were enrolled and their views on the use of generative AI. Students in art and design programmes were strongly critical of its use on ethical and moral grounds, citing sustainability and copyright concerns as key motivations. These students expressed a preference for working more analogically. One teacher noted that the students seemed fatigued by screen-based work, having grown up with constant digital exposure both at home and in school. According to the informants, students in technology and media programmes were generally more positively inclined towards generative AI. However, even within these groups, several students preferred to maintain full control over all aspects of the creative process.

4.4.4 Challenges in evaluating student work

The teachers' assessment practices have been affected by the introduction of text-to-image generative AI. Some informants described clear instances of attempted misconduct, such as students presenting AI-generated material as their own photographs, as well as uncertainty among assessors regarding what constitutes acceptable and unacceptable use of text-to-image generation. There also appeared to be variation in experiences related to examinations and in perspectives on future priorities—whether greater emphasis should be placed on assessing the final product or the underlying creative process.

4.4.5 Strict regulations from the school

Three of the teachers work in schools where the county municipality has implemented strict regulations on which generative tools may be used, limiting access to a single language model that performs poorly and is not designed for image generation. One teacher described these restrictions as contradictory to the official curriculum and subject framework, which emphasise the importance of engaging with new technologies and developing technological understanding as part of the subject's learning outcomes.

4.4.6 Lack of guidelines and information

The teachers reported receiving limited information from their employer or public authorities and described spending significant personal time familiarising themselves with new technologies. One teacher called for clearer guidelines regarding their use.

4.4.7 Lack of knowledge and understanding from peers and leadership

The teachers agreed that school leadership lacks understanding and knowledge about the use of text-to-image generative AI within their subject areas and professional fields. As stated in the word of one informant:

> So it becomes a kind of general talk where everything related to AI is basically a no. […] there is an understanding that AI is just ChatGPT and that it basically has to do with text, so they don't know that it both exists within our software and that it is used for creative

processes—because I don't think they themselves engage in very many creative processes. (Informant 2).

One teacher also emphasizes a lack of interest from colleagues.

5 DISCUSSION

Our informants primarily used text-to-image generators during the ideation phase and early stages of student projects. Previous studies have shown that such tools can positively influence idea variation and help avoiding design fixation (Chen et al., 2019; Hutson & Lang, 2023; Ko et al., 2023; Liu et al., 2023; Vartiainen & Tedre, 2023). Furthermore, the teachers in our study described generating raw materials to be edited and incorporated into final outcomes, which we have not found in previous studies.

The teachers were conscious to develop the students' critical understanding of AI and how visual outputs in society is affected by it, much in line with what is described as AI literacy (Kong et al., 2024; Chiu, 2025; OECD, 2025). However, they were divided on whether students need to master the technology itself. Some considered prompt engineering a skill worth cultivating, echoing Oppenlaender et al. (2025), while others regarded visual literacy as sufficient. Taken together, the teachers' perspectives suggest that visual literacy is viewed as a component of AI literacy—consistent with Chiu's (2025) argument—particularly in relation to evaluating generated images.

The study also illustrates how teachers are balancing between demands to prepare the students for a future with AI, with hindrances in form of negative perceptions and limited access to functional technology. The teachers experience either overly restrictive or entirely absent guidelines, both attributed to an unfamiliarity with generative technology within art, design, and media contexts. Some also encountered student resistance to using the tools. Similar ambivalence has been reported in previous studies (Heaton et al., 2024; Vartiainen & Tedre, 2023; Vartiainen et al., 2023).

Recognising the importance of the emerging technology, the teachers are caught between several conflicting considerations in their efforts to prepare students accordingly. If schools and teachers are to be able to prepare students for contemporary and future working life and fulfil the requirements of the curriculum, they must have access to functional tools and up-to-date knowledge. It is the responsibility of schools and public authorities to facilitate this.

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